Shared multi-view encoder (identical, train & inference)

5 views per image **[NEW: strong augmentation]**
1 global (288x288) + 4 local corner crops (64% width)
orientation-corrected first (`fix_orientation=true`)
global view: RandomResizedCrop scale **(0.3, 1.0)** (was 0.65) + **TrivialAugmentWide** (new) + ColorJitter + rot 5° + RandomErasing
no horizontal flip — it would mirror driving side, signage and text

$5 \times 3 \times 288 \times 288$

NEW: MobileNetV4-small backbone – SHARED across all 5 views
`mobilenetv4_conv_small.e2400_r224_in1k, num_classes=0`
2,493,024 params (was EfficientNet-B0's 4,007,548)
⇒ **frees 1.51M params for the cell heads**
from ImageNet init, 120 epochs, cosine

5×1280

Per-view features
5×1280 (1 global, 4 local)

Scene feature → full-rank classification/offset heads

5×1280

Fuse scene feature
$0.5 \cdot \text{global} + \text{mean}(4 \text{ local})$

1280

Dropout p=0.3
→ fused feature

1280 | 1280 | 1280

Country head
Linear $1280 \to 12$
15,372 params

Country CE loss
weight 1.0

NEW: full-rank cell heads k=0..3
$4 \times$ Linear $1280 \to 368$ **(direct, no trunk)**
E3e routed these through a shared Linear$(1280 \to 224)$+GELU trunk to fit the budget; here the freed backbone params pay for the full-rank heads instead — **1,885,632 params, 40% of the entire 5M budget**. 368 cells/partition, K=4 partitions.
raw **unmasked** logits

Haversine-smoothed cell loss ($\times$4) — kept from E3e
target $\propto \text{softmax}(-\text{hav}(\text{centroid}_i, \text{truth})/\tau)$, τ=45 km
over that image's country's cells (PIGEON eq. 2–3), masked to **true** country (teacher forcing)

Geographic-distribution loss
pairwise embedding cosine vs. smoothed pairwise haversine distance (scale 250 km); weight 0.2, temperature 0.1

View-consistency loss
cosine align all 5 per-view embeddings to the fused one; weight 0.1

NEW: supervised geo-contrastive loss REMOVED
`contrastive_weight` 0.2 → 0.0
the embedding is now shaped only by the geo-distribution and view-consistency terms

From ImageNet init, **120 epochs** (E3e: 55), batch 32, backbone_lr $1e{-4}$ / head_lr $1e{-3}$, wd $1e{-3}$, seed 0, EMA off (`ema_decay=0.0`), ImageNet mean/std, crop_pct 0.875.
Best epoch 114/120: 126.04 km (fold-0 val median).
country acc 0.781, cell top-1 acc 0.141 (368-way, 4 partitions)

Offset head
Linear $1280 \to 2$
(trained, but the tuned decoder sets o=0: no help)

Offset regression loss
weight 0.5

Embedding branch

5×1280

Embedding head (per view)
Linear $1280 \to 256$, L2-normalize
(256-d again; E3e had shrunk it to 224 for budget)

5×256

Fuse & re-normalize
$0.5 \cdot \text{emb}_{\text{global}} + \text{mean}(4 \text{ local emb})$

Fused query embedding
256-d, L2-normalized

B. Inference-time retrieval decode

Gallery (built once, offline, frozen)
every fold-*training* image's country, 4×cell_id, fused & local embeddings, unit-vector coords

Hierarchy score
$\log P(\text{country}) + \frac{1}{4}\sum_{k=0}^{3} \log P(\text{cell}_k \mid \text{country})$
(country-masked conditional log-softmax per partition)

Similarity score
$(1-\ell) \cdot \cos(q_{\text{fused}}, g_{\text{fused}}) + \ell \cdot \max_v \cos(q^v_{\text{local}}, g_{\text{local}})$
(local_weight ℓ; **best fold-0**: ℓ=0)

Combined score
$\text{hierarchy} + w \cdot \text{similarity}$
retrieval_weight w; **best**: w=16

Spherical mode decode
top-m by score → pairwise great-circle distances → highest-mass coherent cluster within radius r (hard kernel) → score-weighted spherical mean
best: m=50, r=200 km, offset off

Predicted (lat, lng)
decimal degrees

`mode_w16_l0_r200`: **126.04 km**
mean 413.8 km, <200 km 0.603, <750 km 0.835
cell-argmax decode, no retrieval: 146.06 km
(train-time grid only — no wider sweep was run)

A. Training-time losses